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Experiment 4: Artificial Neural Network using Backpropagation

Aim:

To build an **Artificial Neural Network** using the **Backpropagation algorithm** and test it using a dataset.

Python Program:

```
import numpy as np


X = np.array([
    [0, 0],
    [0, 1],
    [1, 0],
    [1, 1]
])


y = np.array([
    [0],
    [1],
    [1],
    [0]
])


def sigmoid(x):
    return 1 / (1 + np.exp(-x))
```

```
def
    sigmoid_derivative(x):
    return x * (1 - x)

np.random.seed(1)

w_hidden = np.random.uniform(size=(2, 2))
b_hidden = np.random.uniform(size=(1, 2))
w_output = np.random.uniform(size=(2, 1))
b_output = np.random.uniform(size=(1, 1))

learning_rate = 0.5
epochs = 10000

for i in range(epochs):
    hidden_input = np.dot(X, w_hidden) + b_hidden
    hidden_output = sigmoid(hidden_input)

    final_input = np.dot(hidden_output, w_output) + b_output
    predicted_output = sigmoid(final_input)

    error = y - predicted_output

    d_output = error * sigmoid_derivative(predicted_output)
    error_hidden = np.dot(d_output, w_output.T)
    d_hidden = error_hidden * sigmoid_derivative(hidden_output)

    w_output += np.dot(hidden_output.T, d_output) * learning_rate
```

```
    b_output += np.sum(d_output, axis=0, keepdims=True) * learning_rate


    w_hidden += np.dot(X.T, d_hidden) * learning_rate
    b_hidden += np.sum(d_hidden, axis=0, keepdims=True) * learning_rate


print("Predicted Output:")
print(np.round(predicted_output, 4))


print("\nClassified Output:")
print((predicted_output > 0.5).astype(int))
```

Output:

```
print((predicted_output

Predicted Output:
[[0.0193]
 [0.9833]
 [0.9834]
 [0.0173]]

Classified Output:
[[0]
 [1]
 [1]
 [0]]
```